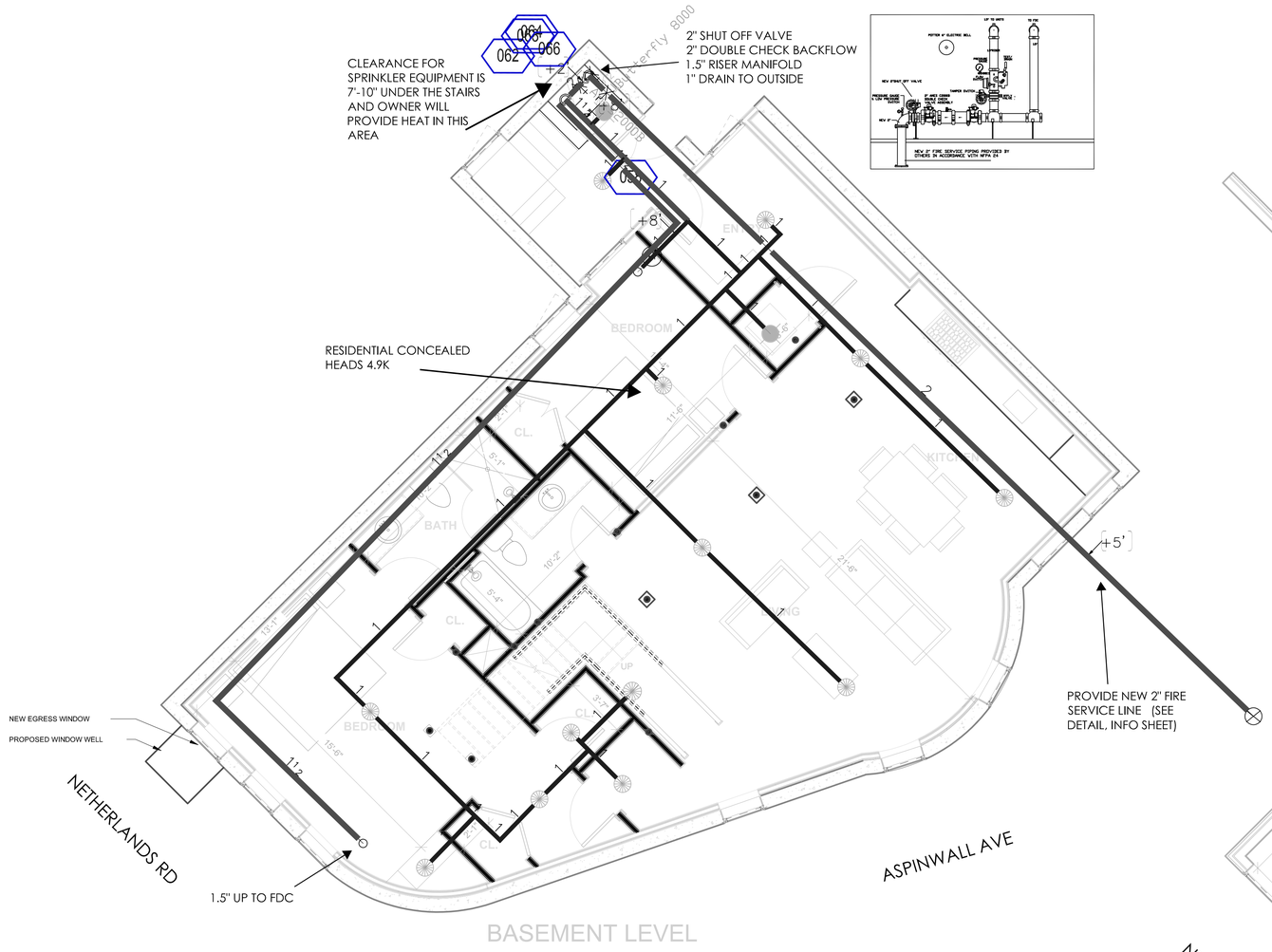
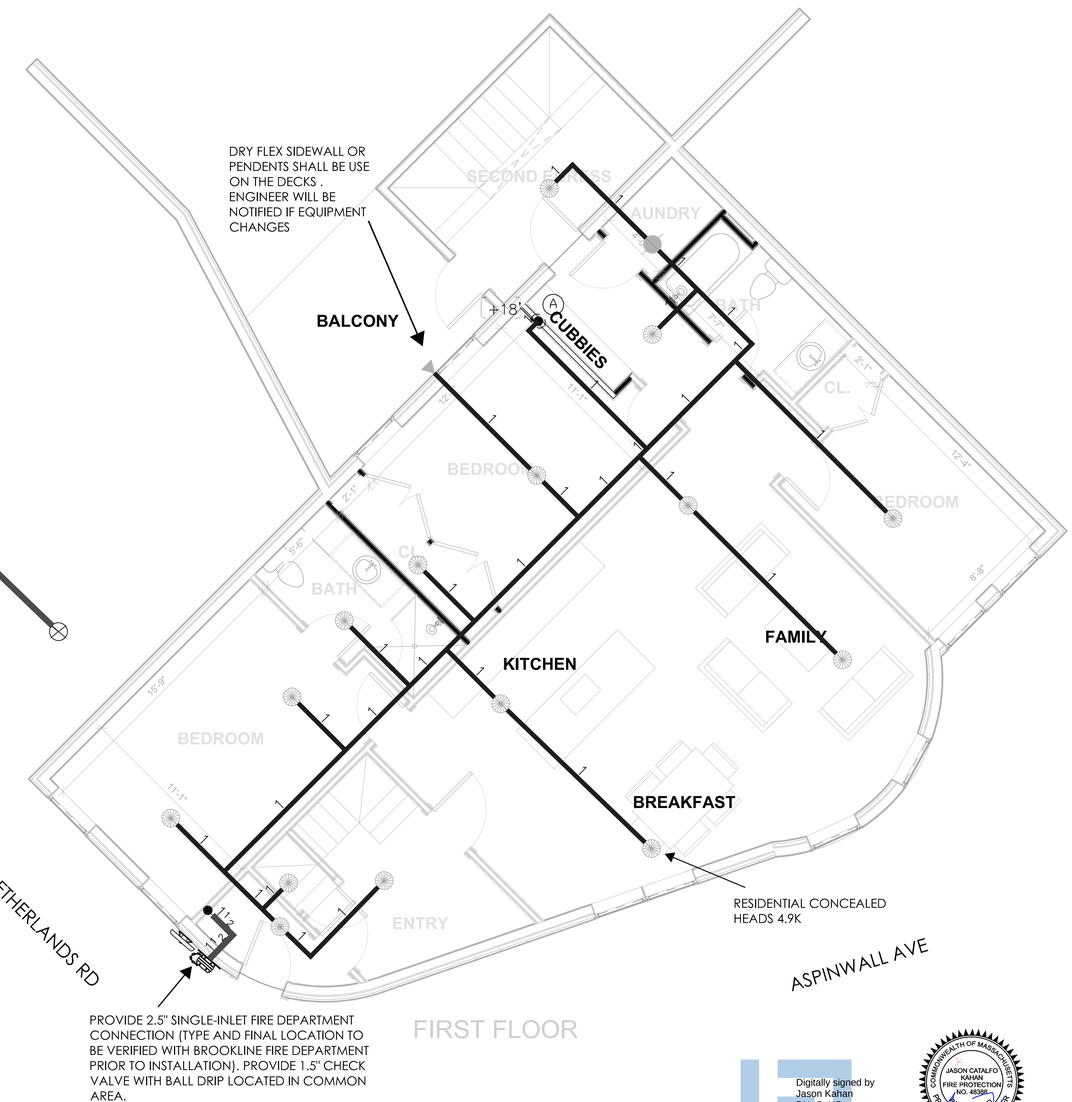




FIRE HYDRANT IS LOCATED DIRECTLY ACROSS THE STREET OF 252 ASPINWALL AVE

Jason Kahan

Digitally signed by Jason Kahan
DN: C=US, E=jk@jasonkahan.com, CN=Jason Kahan
Date: 2025.10.07 15:48:03-04'00'

COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO KAHAN
FIRE PROTECTION
NO. 48389
REGISTERED PROFESSIONAL ENGINEER
10/07/2025

NOTES

NB DESIGN LLC
109 TENNYSON DR
MANCHESTER,NH 03104

NB DESIGN LLC

REVISIONS		
DATE	DESCRIPTION	BY

SPRINKLER SYMBOLS

FIRE PROTECTION
RESIDENTIAL SYSTEM
NFPA 13R

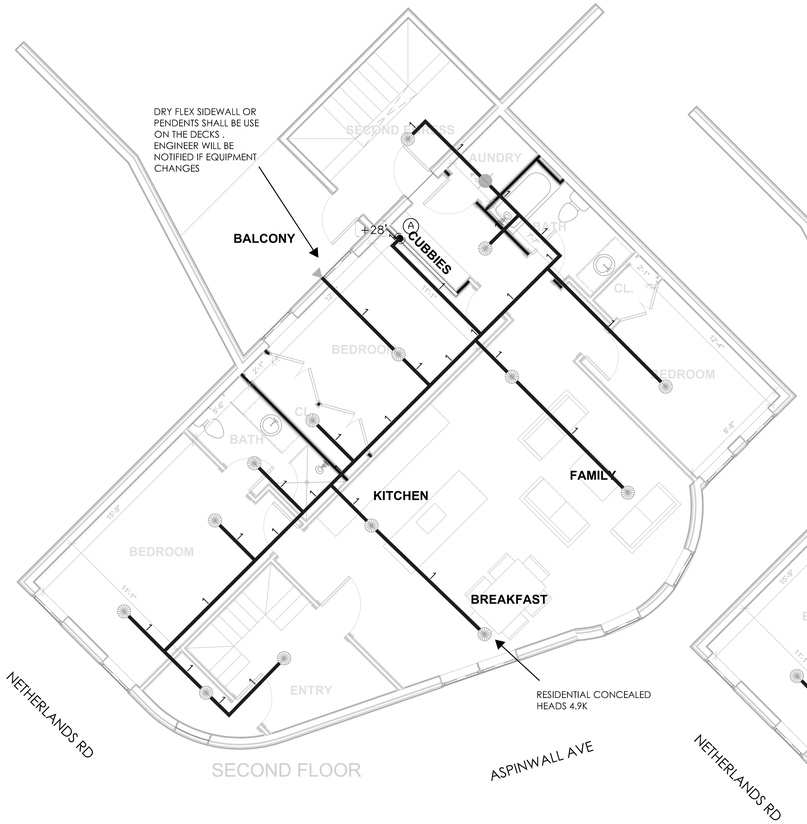
RESIDENTIAL MULTI UNIT
252 ASPINWALL AVE
BROOKLINE, MA 02445

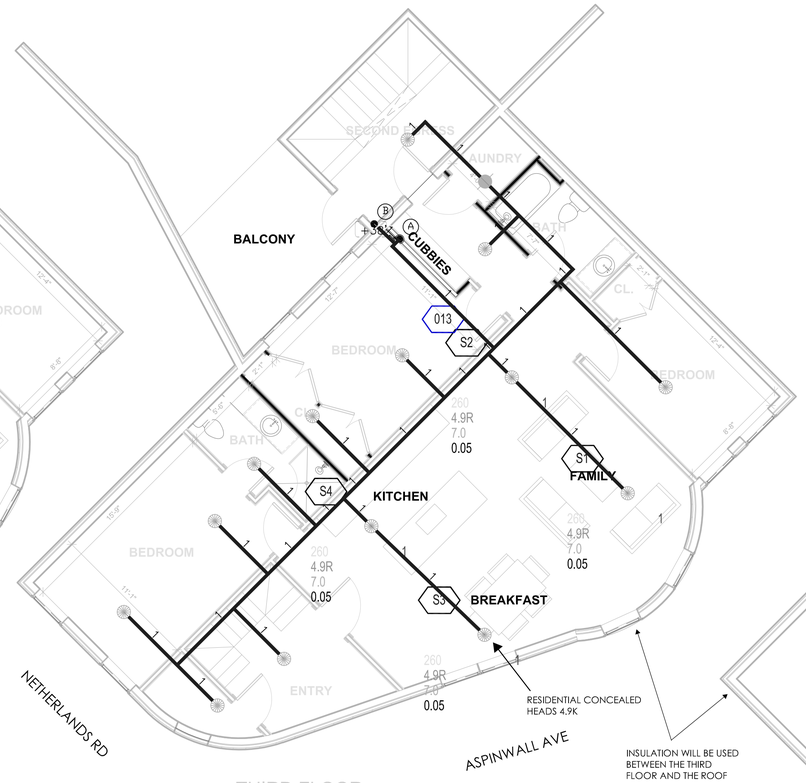
CONTRACTOR:

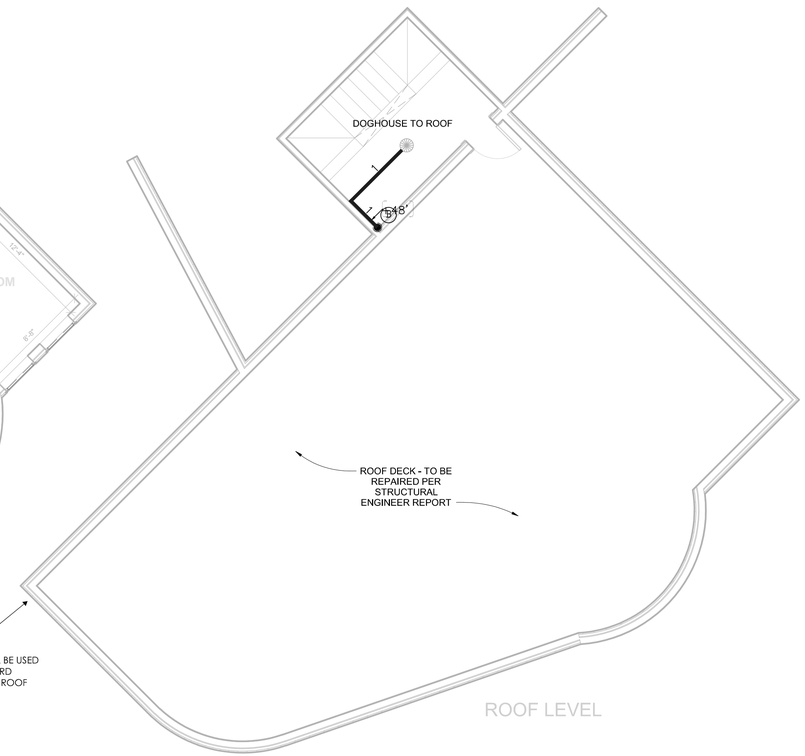
PIPING PLAN BASE-1ST

PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	NB
SCALE	1/4" = 1'-0"
DATE	10/6/25
REVISED	
PLOTTED	

FP 1







Calculation results for Design Area 1 – TOP FLOOR

This system as shown on _____ company print no _____ dated 8/11/25
for RESIDENTIAL MULTI UNIT at 252 ASPINWALL AVE
contract no _____ is designed to discharge at a rate of 0.05 gpm/ft² (L/min/m²) of floor area over a maximum area of 510.64 ft² when supplied with water at a rate of 55.8 gpm at 68.3 psi at the base of the riser. Hose stream allowance of _____ is included in the above.

Occupancy classification: LH
Commodity classification: _____
Maximum storage height: _____
Storage arrangement: _____

Number of heads flowing: 4
System Type: Wet
Maximum velocity: 16.92 ft/s

Flow from In-Rack sprinklers:	0 gpm
Flow from Overhead sprinklers:	55.8 gpm
Flow from Inside Hoses:	0 gpm
Flow from Outside Hoses:	0 gpm
Other fixed flows:	0 gpm
Total flow in system piping:	55.8 gpm
Additional flow at/beyond source:	100 gpm
Total of all flows:	155.8 gpm

Pressure Required at Source:	68.3 psi
Pressure Available at Source:	110.7 psi
Surplus Pressure at Source:	42.4 psi

Design Area 1	Wet System
TOP FLOOR	
Demand Calculations using Hazen-Williams Method	
Occupancy Classification:	LH
Design Area Density:	0.05
Additional Outside Hose:	100
Design Area Size:	510.64
Notes:	

COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO
KAHAN
FIRE PROTECTION
NO. 45356
REGISTERED
PROFESSIONAL ENGINEER
10/07/2025

N
W E
S

NOTES

NB DESIGN LLC
109 TENNYSON DR
MANCHESTER,NH 03104

REVISIONS		
DATE	DESCRIPTION	BY

SPRINKLER SYMBOLS

SPRINKCAD

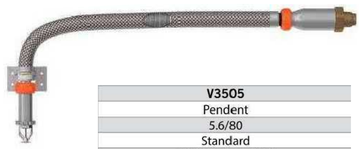
FIRE PROTECTION
RESIDENTIAL SYSTEM
NFPA 13R

RESIDENTIAL MULTI UNIT
252 ASPINWALL AVE
BROOKLINE, MA 02445

CONTRACTOR:

PIPING PLAN
2ND-ROOF

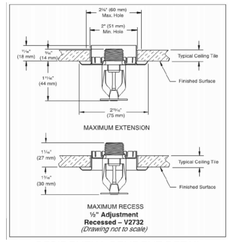
PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	NB
SCALE	3/16" = 1'-0"
DATE	10/6/25
REVISED	
PLOTTED	FP 2



V3505
Pendent
5.6/80
Standard
1' NPT/25mm BSPT
175 psi (1200 kPa)
Sleeve and Skirt
38"/965mm, 50"/1270mm and 58"/1473mm

DRY-FLEX PENDENT DETAIL
SCALE: NOT TO SCALE

STANDARD RESPONSE SLEEVE AND SKIRT HORIZONTAL SIDEWALL SPRINKLER	
SIN	V3509
ORIENTATION	Horizontal Sidewall
NOMINAL K-FACTOR/METRIC K-FACTOR	5.6/80
RESPONSE	Standard
CONNECTION	1" NPT/25mm BSPT
MAX. WORKING PRESSURE	175 psi (1200 kPa)
ESCUTCHEON	Sleeve and Skirt
LENGTHS	38"/965 mm, 50"/1270 mm and 58"/1473 mm

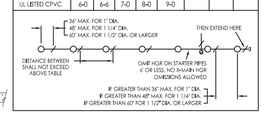


RESIDENTIAL PENDENT DETAIL
SCALE: NOT TO SCALE

Job:	RESIDENTIAL MULTI UNIT		
Address:	252 ASPINWALL AVE		
City:	BROOKLINE	State: MA	Zip/Postal Code: 02445
Contract #:		Date:	8/11/25
Contractor:		Designer:	NB
Address:			
City:		State:	Zip/Postal Code:
Phone:		Fax:	Email:
Approving Authority:			
Design Defaults		Standards:	NFPA13R
Default Sprig Size:	1	Default Drop Size:	1
Default Sprig Material:	CPVC	Default Drop Material:	CPVC
Default Sprig Elevation:	0	Default Drop Elevation:	0

MAX. HANGER SPACING

PIPE TYPE	NOMINAL PIPE SIZE (IN)							
	1	1¼"	1½"	2	2½"	4	6	8
SCH 10,40 STEEL	12-0	12-0	15-0	15-0	15-0	15-0	15-0	15-0
UL LISTED CPVC	6-0	6-6	7-0	8-0	9-0			



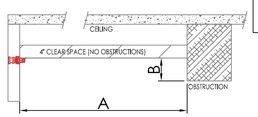


FIGURE 8.10.7.1.3
POSITIONING OF SPRINKLERS TO AVOID OBSTRUCTION TO DISCHARGE

TABLE 8.10.7.1.3 POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE	
DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF DEFLECTOR ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 8 FT	NOT ALLOWED
8'-0" TO LESS THAN 10'	0'-1"
10' TO LESS THAN 11'	0'-2"
11' TO LESS THAN 12'	0'-3"
12' TO LESS THAN 13'	0'-4"
13' TO LESS THAN 14'	0'-6"
14' TO LESS THAN 15'	0'-7"
15' TO LESS THAN 16'	0'-9"
16' TO LESS THAN 17'	0'-11"
17' OR GREATER	1'-2"

NFPA 13 TABLE 8.10.7.1.3 + FIGURE 8.10.7.1.3
RESIDENTIAL SIDEWALL SPRINKLERS

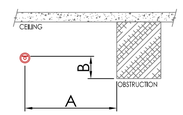


FIGURE 8.10.7.1.4 (B)
POSITIONING OF SPRINKLERS TO AVOID OBSTRUCTION TO DISCHARGE

TABLE 8.10.7.1.4 POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE	
DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF DEFLECTOR ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1'-6"	0'-0"
1'-6" TO LESS THAN 3'-0"	0'-1"
3' TO LESS THAN 4'	0'-3"
4' TO LESS THAN 4'-6"	0'-5"
4'-6" TO LESS THAN 6'	0'-7"
6' TO LESS THAN 6'-6"	0'-9"
6'-6" TO LESS THAN 7'	0'-11"
7' TO LESS THAN 7'-6"	1'-2"

NFPA 13 TABLE 8.10.7.1.4 + FIGURE 8.10.7.1.4 (B)
STANDARD SIDEWALL SPRINKLERS

CLOSETS (TYPICAL)

EXCEPT WHERE SPECIFIED IN SECTION 6.6.4 (NFPA 13D,13R, 2013 EDITION), SPRINKLERS SHALL NOT BE REQUIRED IN CLOTHES CLOSETS, LINEN CLOSETS, AND PANTRIES THAT MEET ALL OF THE FOLLOWING CONDITIONS:

1) THE AREA OF THE SPACE DOES NOT EXCEED 24 SF
2) THE LEAST DIMENSION DOES NOT EXCEED 3 FT
3) THE WALLS AND CEILINGS ARE SURFACED WITH NONCOMBUSTIBLE OR LIMITED-COMBUSTIBLE MATERIALS AS DEFINED IN NFPA 220.

NOTE: WHEN MECHANICAL EQUIPMENT OR LAUNDRY MACHINES ARE PLACED IN THE CLOSET, THE CLOSET IS NO LONGER CONSIDERED A CLOTHES CLOSET, LINEN CLOSET, OR PANTRY, SO THE EXCEPTION OF 6.6.3 IS NO LONGER VALID AND SPRINKLERS MUST BE INSTALLED.

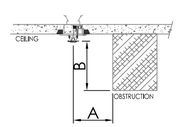


FIGURE 8.10.6.1.2(A)
POSITIONING OF SPRINKLERS TO AVOID OBSTRUCTION TO DISCHARGE

NFPA 13 TABLE 8.10.6.1.2 + FIGURE 8.10.6.1.2(A)
RESIDENTIAL PENDENT AND UPRIGHT SPRAY SPRINKLERS

TABLE 8.10.6.1.2 POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE	
DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF DEFLECTOR ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1 FT	0
1'-0" TO LESS THAN 1'-6"	0'-0"
1'-6" TO LESS THAN 2'-0"	0'-1"
2'-0" TO LESS THAN 2'-6"	0'-1"
2'-6" TO LESS THAN 3'-0"	0'-1"
3'-0" TO LESS THAN 3'-6"	0'-3"
3'-6" TO LESS THAN 4'-0"	0'-3"
4'-0" TO LESS THAN 4'-6"	0'-5"
4'-6" TO LESS THAN 5'-0"	0'-7"
5'-0" TO LESS THAN 5'-6"	0'-7"
5'-6" TO LESS THAN 6'-0"	0'-7"
6'-0" TO LESS THAN 6'-6"	0'-9"
6'-6" TO LESS THAN 7'-0"	0'-11"
7'-0" AND GREATER	1'-2"

TABLE 8.3.2.5(c) TEMPERATURE RATINGS OF SPRINKLERS IN SPECIFIED RESIDENTIAL AREAS		
HEAT SOURCE	MINIMUM DISTANCE FROM EDGE OF SOURCE TO ORDINARY TEMPERATURE SPRINKLER (INCHES)	MINIMUM DISTANCE FROM EDGE OF SOURCE TO INTERMEDIATE TEMPERATURE SPRINKLER (INCHES)
SIDE OF OPEN OR RECESSED FIREPLACE	36	12
FRONT OF RECESSED FIREPLACE	60	36
KITCHEN RANGE	18	9
WALL OVEN	18	9
SIDE OF CEILING OR WALL MOUNTED HOT AIR DIFFUSER	24	12
FRONT OF WALL MOUNTED HOT AIR DIFFUSER	36	18
HOT WATER HEATER OR FURNACE	6	3
LIGHT FIXTURE: DW=250W	6	3
LIGHT FIXTURE: 250W=499W	12	6

FIRE PROTECTION NOTES:

THE PURPOSE OF THIS FIRE PROTECTION DRAWING AND THE ASSOCIATED FIRE PROTECTION DESIGN NARRATIVE IS TO INDICATE THE PROPOSED RESIDENTIAL SPRINKLER SYSTEM TO BE INSTALLED WITHIN THE RENOVATED MULTI-UNIT BUILDING LOCATED AT 252 ASPINWALL AVE BROOKLINE, MASSACHUSETTS.

THIS BUILDING CONSISTS OF A BASEMENT LEVEL, FIRST LEVEL, SECOND LEVEL AND A THIRD LEVEL AS INDICATED ON THE ASSOCIATED ARCHITECTURAL DRAWINGS.

THIS RESIDENTIAL SPRINKLER SYSTEM HAS BEEN DESIGNED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13R (2019 EDITION) FOR A RESIDENTIAL BUILDING UP TO 4 STORIES IN HEIGHT .

THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13R (2019 EDITION), MASSACHUSETTS STATE BUILDING CODE (780 CMR, 10TH EDITION) AND BROOKLINE FIRE DEPARTMENT REQUIREMENTS.

THIS SYSTEM WILL BE SUPPLIED BY A NEW 2" FIRE SERVICE, TAPPED OFF THE EXISTING WATER MAIN ON ASPINWALL AVE. THE 2" SERVICE SHALL BE DESIGNED, INSTALLED, FLUSHED AND TESTED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 24 BY A LICENSED UNDERGROUND CONTRACTOR AND WILL ENTER THE BASEMENT LEVEL AS INDICATED ON THIS DRAWING.

THE SYSTEM HAS BEEN HYDRAULICALLY CALCULATED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13R, INCLUDING THE FOUR HYDRAULICALLY MOST DEMANDING HEADS IN A SINGLE COMPARTMENT BASED ON THE REQUIREMENTS OF THE SPECIFIC SPRINKLER HEAD AND THE SPACING USED IN THIS DESIGN (16'x16' REQUIRING A MINIMUM OF 13 GPM @ 7 PSI). REMOTE AREAS, DENSITIES AND HOSE STREAM ALLOWANCES ARE INDICATED ON THIS DRAWING ALONG WITH SYSTEM DEMANDS AT CONNECTION TO STREET AND CALCULATION RESULTS HAVE BEEN COMPARED TO RECENT HYDRANT FLOW TEST INFORMATION OBTAINED FROM THE BROOKLINE WATER & SEWER DEPARTMENT. CALCULATIONS HAVE BEEN SUBMITTED WITH THIS DRAWING TO THE BROOKLINE FIRE DEPARTMENT FOR REVIEW.

SPRINKLER HEAD LOCATIONS HAVE NOT BEEN COORDINATED WITH CEILING-MOUNTED FIXTURES. INSTALLING SPRINKLER CONTRACTOR SHALL COORDINATE WITH ELECTRICIAN AND LOCATE SPRINKLERS AT LEAST 36" FROM THE CENTER OF ANY CEILING-MOUNTED FIXTURE, IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13 WHILE MAINTAINING THE MAXIMUM DISTANCES FROM WALLS INDICATED ON THIS DESIGN (8-FT). ADDITIONAL SPRINKLERS REQUIRED DUE TO INABILITY TO MEET OBSTRUCTION CRITERIA SHALL BE AT THE OWNERS EXPENSE.

RESIDENTIAL
FLAT PLATE CONCEALED PENDENT

SPRINKLER CONTRACTOR SHALL TAKE PRECAUTIONS WHEN INSTALLING SPRINKLER PIPING IN JOISTS ON TOP FLOOR. GENERAL CONTRACTOR SHALL BE REQUIRED TO INSTALL SUFFICIENT INSULATION TO MAINTAIN 40-DEGREES IN ALL AREAS WHERE SPRINKLER PIPING AND HEADS ARE INSTALLED

INSTALLATION NOTES:

ALL WORK SHALL BE PERFORMED BY A MASSACHUSETTS LICENSED SPRINKLER CONTRACTOR. THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13R (2019 EDITION), MASSACHUSETTS STATE BUILDING CODE (10TH EDITION) AND THE BROOKLINE FIRE DEPARTMENTS.

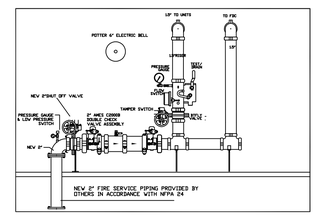
THE ARCHITECTURAL BACKGROUND OF BUILDING MAY DIFFER SLIGHTLY FROM ACTUAL LAYOUT. DRAWINGS ARE NOT INTENDED TO SHOW ALL OFFSETS AND PIPING ELEVATION CHANGES. CONTRACTOR SHALL FIELD VERIFY ALL MEASUREMENTS PRIOR TO FABRICATION.

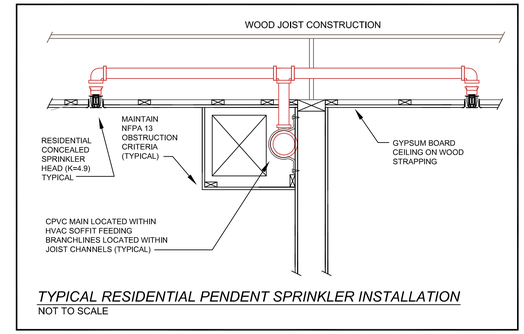
CONTRACTOR SHALL HYDROSTATICALLY TEST ALL SPRINKLER PIPING AT 200 PSI FOR 2 HOURS AND IS RESPONSIBLE FOR THE COMPLETION OF ALL ABOVE GROUND TEST CERTIFICATES, SUPPLIED TO THE OWNER.

ALL PIPING INSTALLED THROUGHOUT THE RESIDENTIAL AREAS OF THE BUILDING SHALL BE UL LISTED CPVC SPRINKLER PIPING.ALL PIPING SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13R (2019 EDITION) AND ALL MANUFACTURERS INSTALLATION RECOMMENDATIONS. ALL PIPING SHALL BE PITCHED TO DRAIN WITH LOW-POINT DRAINS AT SECTIONS OF PIPING SUBJECT TO WATER TRAPPING. SPRINKLER CONTRACTOR SHALL PROVIDE PROTECTIVE PLATES WHERE CPVC PIPING IS RUN THROUGH STUDS TO PREVENT PUNCTURING OF THE SPRINKLER PIPING DURING DRYWALL INSTALLATION AS REQUIRED BY NFPA STANDARDS.

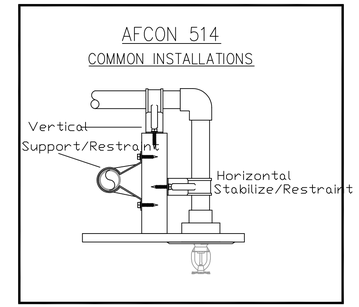
ALL SPRINKLER HEADS WITHIN RESIDENTIAL AREAS OF THE BUILDING SHALL BE RESIDENTIAL PENDENT SPRINKLERS. ALL HEADS SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS AND THE REQUIREMENTS OF NFPA 13R (2019 EDITION).

THE BUILDING OWNER IS RESPONSIBLE FOR MAINTAINING THIS SPRINKLER SYSTEM IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 25, INCLUDING THE PROVISION OF HEAT IN ALL AREAS CONTAINING SPRINKLER PIPING AND HEADS TO PREVENT PIPE FROM FREEZING. THE ENGINEER OF RECORD TAKE NO RESPONSIBILITY FOR ANY DAMAGES CAUSED BY FREEZE UPS.





TYPICAL RESIDENTIAL PENDENT SPRINKLER INSTALLATION
NOT TO SCALE

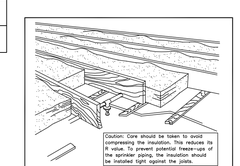


SYM	CNT	POSITION	FINISH	TEMP	K	NPT	SIN	MFG.	MODEL#
	54	PEND	BRASS	155	4.90	1/2"	TY2234	Tyco	LFII
	5	PEND	BRASS	200	5.60	1/2"	TY3551	Tyco	RFLL
	2	DRY SIDE	CHROME	165	5.60	1/2"	TY3322	Tyco	TY-FRL

UL LISTED CPVC SPRINKLER PIPING

FREEZE PROTECTION

THE BUILDING OWNER IS RESPONSIBLE FOR PROVIDING HEAT IN ALL AREAS CONTAINING SPRINKLER PIPING AND HEADS TO PREVENT PIPE FROM FREEZING. ANY AREAS THAT RAISE CONCERN WITH REGARD TO FREEZING POTENTIAL SHALL BE BROUGHT TO THE ENGINEER'S ATTENTION, IN WRITING, PRIOR TO INSTALLATION. THE ENGINEER OF RECORD TAKE NO RESPONSIBILITY FOR DAMAGES CAUSED BY FREEZE-UPS OF THE SPRINKLER SYSTEM.



INSULATION PROTECTING SPRINKLER PIPING ON UPPER FLOOR DETAIL
NOT TO SCALE

COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO
KAHAN
FIRE PROTECTION
NO. 48586
REGISTERED PROFESSIONAL ENGINEER
10/07/2025

N W E S

NOTES

NB DESIGN LLC
109 TENNYSON DR
MANCHESTER,NH 03104

REVISIONS		
DATE	DESCRIPTION	BY

SPRINKLER SYMBOLS

FIRE PROTECTION
RESIDENTIAL SYSTEM
NFPA 13R

RESIDENTIAL MULTI UNIT
252 ASPINWALL AVE
BROOKLINE, MA 02445

CONTRACTOR:

PIPING PLAN INFO SHEET	
PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	NB
SCALE	1/4" = 1'-0"
DATE	10/6/25
REVISED	
PLOTTED	FP 3